PRODUCT INFORMATION BULLETIN

DATE: February 2017 (Rev. J)

SUBJECT: Installation and adjustment instructions for the magnetic pick-up used as an RPM sensor for Evolution, Nexus, R, H and G controls.

MODELS AFFECTED: Any liquid-cooled engine driven generator.

PROBLEM: If the magnetic pick-up is not properly adjusted, the magnetic pick-up will be damaged by the rotating flywheel.

WARRANTY: Technical references used during the procedure: WIB07-08-S: magnetic pick-up (MPU) warranty Information sheet.

CORRECTIVE ACTION: This bulletin explains the procedure for setting the magnetic pick-up voltage output to its optimum value for each respective control panel (Evolution, Nexus, R-series, H-100 and G-panel). The technician will need the Generac Magnetic Pickup Test Kit Harness (P/N# 0G41800SRV) and a Digital Volt Ohm Meter (DVOM) to perform this adjustment. The adjustment is performed with the engine running at rated speed.

PROCEDURE:

1. Verify the unit is OFF.
2. Disconnect the negative (-) battery cable from the battery.
3. Turn the fuel supply OFF.
4. With the magnetic pick-up removed, use a flashlight and verify that a flywheel tooth is directly below and centered in the magnetic pick-up hole. Use an appropriate tool on the crankshaft damper mounting bolt to rotate the crankshaft and flywheel in the direction of normal rotation. If access to crankshaft damper mounting bolt is not possible, then reconnect the battery and turn the engine over by toggling the manual switch to align the tooth underneath the hole.
5. Hand tighten the magnetic pick-up until it contacts the top of the flywheel tooth (Figure 1).

WARNING

Do not rotate the alternator by the rotor fan blades. Doing so can cause catastrophic failure of the fan blades and could result in death or serious injury.

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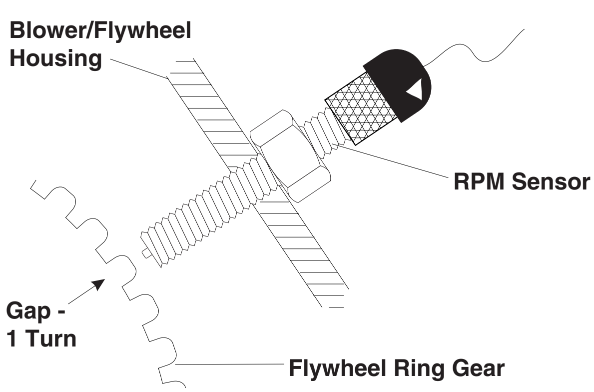

Figure 1

6. Loosen the magnetic pick-up one (1) full turn.
7. Connect the breakout harness (P/N 0G41800SRV) to the magnetic pick-up, in-line with the engine control harness (Figure 2).

Technician Note: The steps below in this procedure must be done with (P/N 0G41800SRV) breakout harness and with a Fluke 87 or equivalent (DVOM). The breakout harness tool can be located in the Guardian warranty part kits level two (2) and level three (3). Contact the Parts Department to order the Magnetic Pick-up Harness Test kit (Figure 2). For information on the Fluke 87 series multi-meter, visit the following website:
http://us.fluke.com

8. Connect the negative (-) battery cable to the battery.

Figure 2

9. Turn the fuel supply ON.
10. Start the unit in MANUAL mode to verify unit starts.
    - If the unit starts, proceed to Step 11.
    - If the unit does not start and if any alarms are displayed on the control board such as;

(Flashing overspeed and/or (RPM) sensor loss), Verify steps one (1) though six (6) were performed correctly. Verify that the magnetic pick-up is producing A/C voltage using the breakout harness and (DVOM) while the engine is cranking. Turn the fuel supply OFF until magnetic pick-up voltage is obtained.

Magnetic pick-up note: The magnetic pick-up may need to be turned in or out a very small amount to get the magnetic pick-up to produce A/C voltage.

11. Set the multimeter to read A/C Voltage and insert leads into wires #79 positive (+) and # 0 ground (-) on the breakout harness.

⚠CAUTION

Avoid damage. Do not overtighten the magnetic pick-up. Contact between the pick-up and flywheel will damage the pick-up.

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12. With the unit running at (Rated Speed: 3600RPM, 1800RPM etc.), slowly adjust the magnetic pick-up to provide the specific voltage for the applicable control board. Verify the part number before adjusting the magnetic pick-up to the appropriate voltage. Examples of the control boards and Rev levels are below:

- The control board and ignition module list using (MPU1) magnetic pick-up, is located on the back of this Product Information Bulletin (PIB).

13. Tighten the magnetic pick-up lock nut to prevent the magnetic pick-up from coming loose during operation. Use Loctite® (blue 232 removable), if available.
14. Verify the magnetic pick-up voltage at rated speed.
15. Turn the unit OFF and remove the breakout harness. Connect the magnetic pick-up connector to the engine harness.

Removal and Inspection of Tip: When the magnetic pick-up is removed, inspect the tip and threads on the body. Clean any filings or debris from the tip and clear threads as needed. Debris on the tip can cause distorted signals to the PCB and ICM which can cause "speed" related shutdowns. This inspection and adjustment procedure should be performed at the annual maintenance visit (Figure 3).

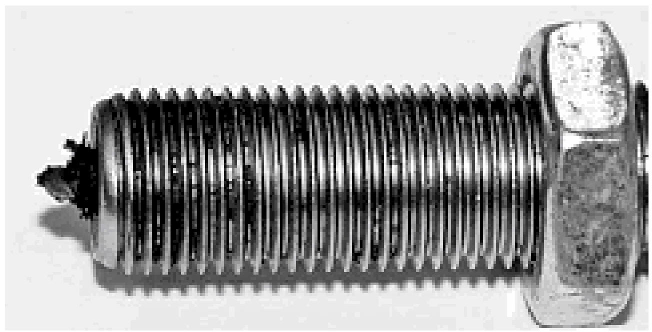

Figure 3

Troubleshooting Tip: The magnetic pick-up P/N 0D2244M, 0G3921, 0G3921A, has a resistance of approximately 1000 Ohms between the small red wire 79 (+) and the small black wire 0 (-) disconnected from the engine harness.

MPU Voltage Setting at Rated Speed as of 07/13/15						
(PCB) Part Number	Type Of Controller	Engine	Rated Speed RPM	*(MPU1) Flywheel Voltage Setting & Tolerance with Generator Running (Also See Note *** below)	**(MPU2) Camshaft Gap Setting	(ICM) Part Number
P/N 0K2267	Evolution	1.5L	3600	7 VAC +/- 0.3 VAC	N/A	N/A
P/N 0K2267	Evolution	2.4L	1800	3 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0K2267	Evolution	2.4L	3600	7 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0K2267	Evolution	5.4L	1800	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0K2267	Evolution	2.3L Diesel	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0K2267	Evolution	2.4L Diesel	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0K2267	Evolution	3.4L Diesel	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	1.5L	3600	7 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	1.6L	3600	7 VAC +/- 0.3 VAC	N/A	N/A

P/N 0H7668	Nexus	2.4L	1800	3 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0H7668	Nexus	2.4L	3600	7 VAC +/- 0.3 VAC	0.15 +/- .001 Gap	N/A
P/N 0H7668	Nexus	4.2L	1800	6 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	4.6L	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	5.4L	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	6.8L	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H1176	R-200C	2.4L	1800	3 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0H1176	R-200C	2.4L	3600	7 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0G8455	R-200B	2.4L	1800	3 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0G8455	R-200B	2.4L	3600	5 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0G8455	R-200B	4.2L	1800	6 VAC +/- 0.3 VAC	N/A	N/A
P/N 0G3958	R-200A	1.6L	1800	14 VAC +/- 0.5 VAC	N/A	N/A
P/N 0G3958	R-200A	1.6L	3600	14 VAC +/- 0.5 VAC	N/A	N/A
P/N 0G3958	R-200A	2.4L	3600	14 VAC +/- 0.5 VAC	.015 +/- .001” Gap	N/A
P/N 0G3958	R-200A	4.2L	1800	6 VAC +/- 0.3 VAC	N/A	N/A
P/N 0G1303	R-200A	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0G1505	R-200A	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0F8710B	R-200	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0F4245	R-100	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0F2815 (Rev A to Rev D)	H-100	2.4L	3600	7 VAC +/- 0.3 VAC	.015 +/- .001” Gap	P/N 0G7021
P/N 0F2815 (Rev A to Rev D)	H-100	2.4L	3600	1 VAC +/- 0.5 VAC	.015 +/- .001” Gap	P/N 0G4649
P/N 0F2815 (Rev A to Rev D)	H-100	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A

P/N 0F2815 (Rev E or Higher)	H-100	All (except for 2.4L)	Any	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0F2815 (Rev E or Higher)	H-100	2.4L	Any	1 VAC +/- 0.5 VAC	N/A	P/N 0G4649
P/N 0E9290 *(5 Amp)	G-100	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0F4649 *(1 Amp)	G-100	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
	E-panel	All	Any	1.5 VAC +/- 0.1 VAC	N/A	N/A

*All voltages are at Rated Speed (Example: 3600rpm, 1800rpm)
*(5 Amp) Sensing (CT) Current Transformer Input
*(1 Amp) Sensing (CT) Current Transformer Input
*(MPU1) Magnetic Pick-up located on Flywheel
**(MPU2) Magnetic Pick-up located on the Camshaft
*** An AC Voltmeter having a suitable frequency response must be used (i.e. Fluke 87)
(N/A = Not Applicable)
(ICM = Ignition Control Module)

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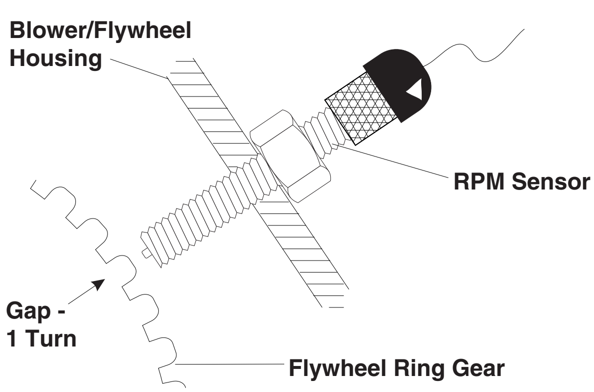

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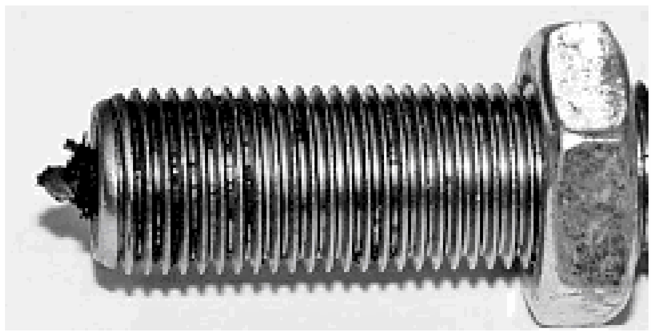

Figure 3

Troubleshooting Tip: The magnetic pick-up P/N 0D2244M, 0G3921, 0G3921A, has a resistance of approximately 1000 Ohms between the small red wire 79 (+) and the small black wire 0 (-) disconnected from the engine harness.

MPU Voltage Setting at Rated Speed as of 07/13/15						
(PCB) Part Number	Type Of Controller	Engine	Rated Speed RPM	*(MPU1) Flywheel Voltage Setting & Tolerance with Generator Running (Also See Note *** below)	**(MPU2) Camshaft Gap Setting	(ICM) Part Number
P/N 0K2267	Evolution	1.5L	3600	7 VAC +/- 0.3 VAC	N/A	N/A
P/N 0K2267	Evolution	2.4L	1800	3 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0K2267	Evolution	2.4L	3600	7 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0K2267	Evolution	5.4L	1800	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0K2267	Evolution	2.3L Diesel	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0K2267	Evolution	2.4L Diesel	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0K2267	Evolution	3.4L Diesel	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	1.5L	3600	7 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	1.6L	3600	7 VAC +/- 0.3 VAC	N/A	N/A

P/N 0H7668	Nexus	2.4L	1800	3 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0H7668	Nexus	2.4L	3600	7 VAC +/- 0.3 VAC	0.15 +/- .001 Gap	N/A
P/N 0H7668	Nexus	4.2L	1800	6 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	4.6L	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	5.4L	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H7668	Nexus	6.8L	All	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0H1176	R-200C	2.4L	1800	3 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0H1176	R-200C	2.4L	3600	7 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0G8455	R-200B	2.4L	1800	3 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0G8455	R-200B	2.4L	3600	5 VAC +/- 0.3 VAC	.015 +/- .001” Gap	N/A
P/N 0G8455	R-200B	4.2L	1800	6 VAC +/- 0.3 VAC	N/A	N/A
P/N 0G3958	R-200A	1.6L	1800	14 VAC +/- 0.5 VAC	N/A	N/A
P/N 0G3958	R-200A	1.6L	3600	14 VAC +/- 0.5 VAC	N/A	N/A
P/N 0G3958	R-200A	2.4L	3600	14 VAC +/- 0.5 VAC	.015 +/- .001” Gap	N/A
P/N 0G3958	R-200A	4.2L	1800	6 VAC +/- 0.3 VAC	N/A	N/A
P/N 0G1303	R-200A	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0G1505	R-200A	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0F8710B	R-200	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0F4245	R-100	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0F2815 (Rev A to Rev D)	H-100	2.4L	3600	7 VAC +/- 0.3 VAC	.015 +/- .001” Gap	P/N 0G7021
P/N 0F2815 (Rev A to Rev D)	H-100	2.4L	3600	1 VAC +/- 0.5 VAC	.015 +/- .001” Gap	P/N 0G4649
P/N 0F2815 (Rev A to Rev D)	H-100	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A

P/N 0F2815 (Rev E or Higher)	H-100	All (except for 2.4L)	Any	5 VAC +/- 0.3 VAC	N/A	N/A
P/N 0F2815 (Rev E or Higher)	H-100	2.4L	Any	1 VAC +/- 0.5 VAC	N/A	P/N 0G4649
P/N 0E9290 *(5 Amp)	G-100	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
P/N 0F4649 *(1 Amp)	G-100	All	Any	1 VAC +/- 0.5 VAC	N/A	N/A
	E-panel	All	Any	1.5 VAC +/- 0.1 VAC	N/A	N/A

*All voltages are at Rated Speed (Example: 3600rpm, 1800rpm)
*(5 Amp) Sensing (CT) Current Transformer Input
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